

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **3D game simulations and environments using Panda3D** for AI training applications, implementing **Python** codebases with integrated **C++** modules to deliver high-performance cross-platform simulations compatible with legacy and contemporary hardware; achieved **40% performance improvement** through optimized 3D graphics pipeline architecture and asset batching.
- Implemented, maintained, and debugged **Python and C++ codebases integrated with Panda3D** across 5+ **simulation projects**, leveraging **GitHub** version control workflows including branching strategies, pull requests, and collaborative code review processes to ensure smooth team collaboration and production-ready deliverables.
- Ensured **cross-platform compatibility and performance optimization** across diverse hardware configurations by profiling 3D assets, implementing dynamic level-of-detail (LOD) rendering, and stress-testing simulation logic across Windows, Linux, and legacy systems; reduced memory footprint by **35%** while maintaining visual fidelity.
- Collaborated with multidisciplinary teams including AI researchers, domain specialists, and QA engineers to transform technical requirements into **3D simulation solutions**, conducting **architecture design reviews**, **brainstorming sessions**, and **team meetings** with clear verbal and written communication of technical decisions and implementation trade-offs.
- Documented architectural decisions, code implementation patterns, and simulation workflows through technical specifications and code comments to support collaborative distributed development; maintained **GitHub wiki and pull request documentation** ensuring team onboarding and knowledge transfer across remote team members.
- Demonstrated exceptional **problem-solving mindset** and **attention to detail** by independently diagnosing and resolving complex 3D rendering bugs, physics simulation edge cases, and hardware compatibility issues in a remote environment; delivered **6+ production simulation builds** on schedule with zero critical defects.

SKILLS

Game Development & 3D Graphics – Panda3D, 3D game simulations, 3D graphics pipelines, simulation logic, 3D asset optimization, cross-platform rendering, LOD (level-of-detail) implementation.

Programming Languages – Python, C++, Java Spring Boot, R.

Version Control & Collaboration – GitHub, Git branching, pull requests, code reviews, distributed Agile environments.

Data & Backend Infrastructure – SQL (PostgreSQL, MySQL, Oracle), REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, CI/CD (Jenkins).

Data Analytics & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing, pandas, NumPy, scikit-learn.

Domain & Compliance – SWIFT messaging (MT/MX), ISO 20022, SEPA, payments processing, ERP integration, GDPR, change-control management.

EDUCATION AND TRAINING

PROJECTS

AI Training Simulation Engine – Multi-Agent 3D Environment (HCL Technologies)

- Architected and developed a **Panda3D-based 3D simulation engine** for AI agent training, implementing **Python** core logic with performance-critical **C++** extensions for physics simulation and collision detection; supported **50+ concurrent AI agents** running at **60 FPS** across Windows, Linux, and legacy hardware platforms.
- Engineered **3D graphics pipeline optimization** including dynamic asset streaming, frustum culling, and texture atlasing to reduce memory consumption by **45%** while maintaining visual realism; profiled and optimized rendering calls achieving **3.2x throughput improvement** on low-spec hardware (Intel HD Graphics legacy systems).
- Implemented robust **GitHub-based CI/CD workflow** with automated testing, build validation, and deployment staging; established branching strategy and pull request review process enabling **8 engineers** to collaborate asynchronously across distributed teams; reduced integration friction by **60%**.
- Designed **cross-platform compatibility layer** abstracting Panda3D platform-specific rendering calls, enabling single codebase deployment across Windows, Linux, and macOS with consistent performance and visual output; validated on **12+ hardware configurations** from modern GPUs to integrated graphics.
- Documented architectural decisions, simulation loop design patterns, and API specifications in **GitHub wiki and technical specifications**; conducted **code review sessions** and **knowledge-transfer meetings** with exceptional written and verbal communication, enabling rapid onboarding of new team members.

SWIFT Messaging Dashboard & Payments Operations Monitoring (HCL Technologies)

- Developed and maintained production-grade **SWIFT messaging dashboard** using **Python** backend services and **REST APIs** processing **50,000+ payment messages daily**, implementing real-time transaction-flow visibility, operational alerting, and **SWIFT MT/MX message format** parsing for banking client production environment.
- Implemented **3-tier monitoring architecture** using **Splunk, CloudWatch, and Dynatrace** to track **SEPA instant payments, cross-border transfers, and ISO 20022 message flows** with **99.95% SLA** compliance; designed alerting rules for P1/P2 incidents reducing mean-time-to-resolution (MTTR) by **35%**.
- Engineered **PostgreSQL** reconciliation workflows capturing payment state transitions, implementing audit trails compliant with **GDPR data protection** and financial regulatory requirements; designed **SQL schema** supporting **1M+ historical transaction records** with sub-second query performance.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)